

ISM Note 4

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1. ISM in ellipicals (continued)

The total mass of ISM in ellipicals is comparable to that in spirals, though the star formation rate is much lower. The great difference and the main reason lies behind low SFR is that **the ISM in ellipicals is multiphase and much hotter**.

For ellipicals, only a small amount of gas is at $10^4 - 10^6$ K, while most of the gas is at temperatures $\geq 10^6$ K. This hot phase produces **strong X-ray emission**, and the typical random motion speed of stars is around 100 km/s. These characteristics are indicative of the presence of hot ISM in ellipical galaxies.

The gas can be heated by **the stellar wind and the AGN feedback**. For the first way, the star moves randomly and sends out the stellar wind and can heat the gas. For another way, the AGN feedback can heat the gas by the radiation and the jet.

2. Radiative process

First of all, we need to define a quantity called Intensity. The intensity is defined by

$$I_\nu = \frac{dE}{dA dt d\Omega d\nu}$$

which is the **energy per unit area per unit time per unit solid angle per unit frequency**, the unit of which is $\text{erg/s/cm}^2/\text{Hz/steradian}$.

Note than here the $d\Omega$ and dA are usually not related since the radiation is not a point source.

Then we can introduce the **flux**, which is defined by

$$F_\nu = \int I_\nu \cos \theta d\Omega$$

and its unit is $\text{erg/s/cm}^2/\text{Hz}$.

Then straightforwardly, we can define the **mean intensity** by

$$J_\nu = \frac{1}{4\pi} \int I_\nu d\Omega$$

Here we don't have the $\cos \theta$ term since we don't need to consider the direction of the radiation.

After that, we can define the **energy density** by

$$u_\nu = \frac{4\pi}{c} J_\nu$$

it is the angular integrated (flux/c), and the unit here is $\text{erg/cm}^3/\text{Hz}$.

The next question is, for different kind of gas, what is the energy density? For simplicity, **we assume the gas is in thermal equilibrium and is ideal gas.**

For a system in thermal equilibrium, there must be a distribution of the energies, to find the energy density we need to find this distribution first. So the next question is, what is the possibility of finding the small system at energy E ? The answer is given by the Boltzmann distribution, which is

$$P(E) \propto e^{-E/kT}$$

while for idealized gas the energy is given by $E = \frac{1}{2}mv^2$, so the velocity distribution is given by

$$P(\vec{v}) \propto e^{-m|\vec{v}|^2/2kT}$$

Then like what we do in thermodynamics, we can introduce the distribution function

$$f(v) = n \left(\frac{m}{2\pi kT} \right)^{3/2} e^{-mv^2/2kT}$$

where $n = \int f(\vec{v}) d^3\vec{v}$.

If we change the variable from vector to scalar, we can get the Maxwell-Boltzmann distribution, which is

$$f(v) = 4\pi v^2 n \left(\frac{m}{2\pi kT} \right)^{3/2} e^{-mv^2/2kT}$$

The mean velocity is given by

$$\langle v \rangle = \int v f(v) dv$$

and the mean kinetic energy is given by

$$\langle E \rangle = \frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} kT$$

The pressure of ideal gas is the force exerted per unit area by particle collisions, which is given by

$$P = \frac{1}{3} n m \langle v^2 \rangle = nkT$$

For a highly relativistic gas, the energy is given by $E = pc$, so the pressure is given by

$$P = \frac{1}{3} n \langle pc \rangle = \frac{1}{3} u$$

where u is the energy density. This type of relativistic particles can be the cosmic rays.

The equation for pressure above actually is **the equation of state**, which can be derived from thermodynamics and is given by

$$P = (\gamma - 1)u$$

where γ is the adiabatic index. For non-relativistic gas, $\gamma = 5/3$, while for relativistic gas, $\gamma = 4/3$.

Radiation (concepts and definitions)

When we say **radiation**, we usually mean the process where photon **interact with matter**. The important thing for radiation process is **the number of photons might not be conserved**: some photon can be absorbed, and some can be re-emitted.

In this picture we assume everything is **in thermal equilibrium** for simplicity. We know from thermodynamics that photon gas follows the Planck distribution (Literally it might be Bose distribution). The number of photon states in 3D reciprocal interval $(\vec{k}, \vec{k} + d\vec{k})$ is given by

$$d^3q = \frac{L^3}{4} \pi k^2 \frac{dk}{(2\pi)^3}$$

where $k = \frac{2\pi q}{l}$ is the wavenumber.

From this we can simply derive the **density of states**, which is given by

$$\frac{d^3q}{L^3} = \frac{8\pi\nu^3}{c^3} d\nu$$

After that, with some steps skipped here we can derive the **energy density of radiation** by

$$u_\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/kT} - 1}$$

and the unit of which is $\text{erg}/\text{cm}^3/\text{Hz}$.

The **flux** of the radiation is then given by

$$F_\nu = \frac{c}{4\pi} u_\nu$$

and the **intensity** is flux divided by the solid angle, which is given by

$$I_\nu = \frac{c}{4\pi} u_\nu$$

Typically we deal with black body, so in astrophysics society we also use the notation $B(\nu, T)$ to represent the intensity.

The total energy density of the radiation is given by

$$u = \int u_\nu d\nu = aT^4$$

The key thing here is **the radiation energy density is proportional to the fourth power of the temperature**, which might should be familiar to readers having learned thermodynamics.

The intensity can be rewritten as

$$B(T) = \frac{c}{4\pi} aT^4$$

and the flux received by the observer is given by

$$F = \frac{ac}{4} T^4$$

We can deduce from the above equations that

$$F = \pi B(T)$$

This is an interesting result since there seems to miss a "4" factor, but this result anyway is correct. The reason is that we can only see **half of the radiation surface**(hemisphere). Also note that the flux here is what we observed, so literally this is the flux **projected onto the line of sight**, so the hemisphere just acts like a circle and therefore we only have a factor of π instead of 4π .

Radiative transfer

Having introduced the necessary concepts, let us now discuss some physics. What is radiative transfer? It is **the process of radiation propagating through the medium**. This might sound a bit abstract, but we can understand it in a more intuitive way: Suppose we shoot out a beam of light into the ISM, then naturally we want to know **what will happen to this beam of light, how it will interact with the ISM, and what we can see after the beam of light goes through the ISM**. This is the question that radiative transfer tries to answer.

Specifically, the radiation can be absorbed, scattered, or emitted by the medium. To give a quantitative description, we need to define some emission coefficient here first.

The **emission coefficient** is defined by

$$j_\nu = \frac{dI_\nu}{dx}$$

which is the change of intensity per unit length, the unit of which is $\text{erg/s/cm}^2/\text{Hz/steradian}$.

If we assume the gas is isortopic, we can derive the **volume emissivity**, which is given by

$$\epsilon_\nu = 4\pi j_\nu$$

this is the most frequently used quantity in astrophysics, which refers to **the energy emitted per unit volume per unit time per unit frequency**. Intuitively speaking, this gives a sense of the **emission strength** of the radiation source, not dependent on the volume, on direction, and can have a frequency spectrum.

Besides, we can also define the **absorption coefficient**, which is defined as

$$dI_\nu = -\kappa_\nu I_\nu dx$$

here κ_ν is the absorption coefficient, which is in unit of cm^{-1} .

From the definition we can see that κ here describes the **fractional absorption or scatter of a radiation beam, per unit length**.

This has a different unit with j_ν , since typically the absorption is proportional to the number density of the photons, thus proportional to the intensity I_ν . We want a physical quantity which would be best if it only depends on the properties of the medium, so simply divided by I_ν is a good choice.

From statistical mechanics, we can have

$$\kappa \sim n\sigma_\nu \sim \frac{1}{\lambda_{mfp}}$$

where n is the number density of the medium and σ_ν is the cross section of the absorption or scattering. When the cross section is large, the absorption is strong, and the mean free

path is short.

With all of these defined, we can finally write down the transfer equation, which is given by

$$\frac{dI_\nu}{dx} = j_\nu - \kappa_\nu I_\nu$$

This includes both the source term and the absorption term, and it describes the change of intensity as the radiation propagates through the medium.

Now we look at one more concepts appeared frequently in astrophysics, that is **the optical depth**, which is defined by

$$\tau_\nu = \int \kappa_\nu dx$$

and from our previous approximation of κ_ν , we can see that

$$\tau_\nu \sim \frac{L}{\lambda_{mfp}}$$

We have optically thin/thick medium, which is defined by

$$\tau \ll 1 \text{ (optically thin)}$$

and

$$\tau \gg 1 \text{ (optically thick)}$$

respectively.

And if we change the variable from x to τ , we can get the transfer equation in terms of optical depth, which is given by

$$\frac{dI_\nu}{d\tau} = S_\nu - I_\nu$$

where $S_\nu = \frac{j_\nu}{\kappa_\nu}$ is called **the source function**.

If we integrate the above equation, we can get

$$I_\nu(\tau) = I_\nu(0)e^{-\tau} + \int S_\nu e^{-(\tau-\tau')} d\tau'$$

The first term is the **attenuation** of the background source, and it describes the absorption and scattering of the background source.

The second term is the contribution from the medium itself, and it describes the emission from the medium.

If we let $I(0) = 0$, e.g. ignoring the background source, we can get

$$I_\nu(\tau) = S_\nu(1 - e^{-\tau})$$

When $\tau \ll 1$, we have

$$I_\nu \sim S_\nu \tau \sim j_\nu L$$

which means **the emission from each element of the medium can be simply added together and get into our eyes.**

When $\tau \gg 1$, we have

$$I_\nu \sim S_\nu$$

which means **the final intensity is just the source function** and it can be quite different from the emissivity. That is because the absorption and scattering inside the optically thick medium is very strong so we cannot see the emissivity directly.

Now we consider the **local thermal equilibrium** (LTE), which means a small volume of gas is internally in thermal equilibrium, even if the entire system is not. Physically, this means the gas is dense enough so that the medium is opaque to its own radiation and optically thick.

In this case, we have

$$S_\nu \sim B_\nu(T)$$

from which we can deduce that

$$j_\nu \sim \kappa_\nu B_\nu(T)$$

This is called as **Kirchhoff's law**, which means the emission coefficient is connected to the absorption coefficient by the black body intensity.

Then we can define the so-called **brightness temperature**, which means the intensity is equal to the black body intensity at the brightness temperature and is defined by

$$I_\nu = B_\nu(T_B)$$

For example, for $\nu \ll k_B T$, we have

$$B_\nu(T) \approx \frac{2k_B T \nu^2}{c^2}$$

From local thermal equilibrium assumption, we have

$$I_\nu \approx B_\nu(T)$$

Then using the solution of the transfer equation, we have

$$I_\nu = S_\nu(1 - e^{-\tau}) \approx B_\nu(T)(1 - e^{-\tau})$$

Plugging in the relation

$$B_\nu(T) \approx \frac{2k_B T \nu^2}{c^2} \propto T$$

we finally get

$$T_B = T(1 - e^{-\tau_\nu})$$

When $\tau \gg 1$, we have $T_B \approx T$, which means the brightness temperature is close to the physical temperature.

When $\tau \ll 1$, we have $T_B \approx T * \tau_\nu$, which means the brightness temperature is much smaller than the physical temperature.

The example for the first case is GMC, where we have $T \approx T_B \approx 10K$, and the example for the second case is ISM, where we have $T_B \approx T * \tau \approx 10^{-2}K$, which is definitely much smaller than the real temperature of the ISM.